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In [2]: # Program 19: IMDb preprocessing - word frequency and review length
import re
import pandas as pd
import matplotlib.pyplot as plt
from collections import Counter
from sklearn.feature_extraction.text import ENGLISH_STOP_WORDS

FILE = r"D:\Datascience-MachineLearning\Datasets\IMDB_Dataset.csv" # <-- change to your file path
TEXT_COL = "review"

df = pd.read_csv(FILE)

def clean(text):
    text = text.lower()
    text = re.sub(r"<.*?>", " ", text) # remove HTML tags
    text = re.sub(r"^[^a-z\s]", "", text) # keep only letters
    return text

df["clean"] = df[TEXT_COL].apply(clean)

# word frequency (without stop words)
words = []
for review in df["clean"]:
    for w in review.split():
        if w not in ENGLISH_STOP_WORDS:
            words.append(w)
print("Top 15 most common words:")
for word, count in Counter(words).most_common(15):
    print(word, ":", count)

# review length (number of words)
df["length"] = df["clean"].apply(lambda t: len(t.split()))
print("\nReview length statistics:")
print(df["length"].describe())

plt.hist(df["length"], bins=40, color="orange", edgecolor="black")
plt.xlabel("Words per review")
plt.ylabel("Number of reviews")

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plt.title("Review Length Distribution")  
plt.show()
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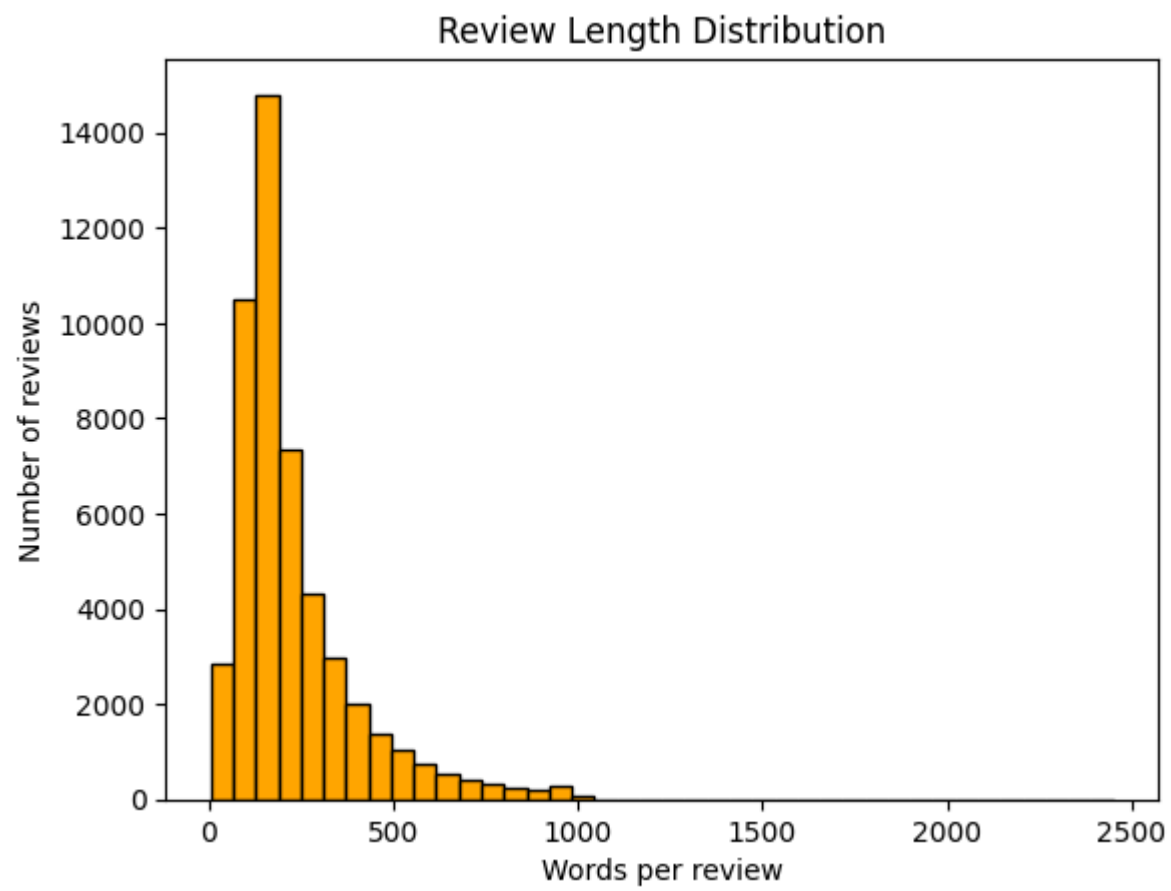
Top 15 most common words:

movie : 85235
film : 76035
like : 39066
just : 34884
good : 28913
time : 23971
really : 23014
story : 22521
bad : 18004
great : 17936
people : 17710
dont : 16976
movies : 15722
films : 15539
way : 15373

Review length statistics:

count	50000.000000
mean	226.850260
std	168.175138
min	4.000000
25%	124.000000
50%	170.000000
75%	275.000000
max	2450.000000

Name: length, dtype: float64



In []: